



Course Code : CSE-3532

Course Title : Tools and Technologies for Internet Programming

Project Title : Sumtasinia Medica : A Web-Based Disease Awareness and Health Information Platform

Team Number : 03

Team Members :

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1 Abstract

Sumatsina is a comprehensive digital healthcare solution designed to act as both a medical encyclopedia and a bridge between patients and healthcare providers. By integrating a vast library of credible disease data with a robust appointment booking system, the platform empowers users to educate themselves and seek professional help simultaneously. It also serves as a repository for hospital datasets, supporting medical research and data-driven healthcare improvements.

2 Introduction

In a world of fragmented health information, Sumatsina provides a "Single Source of Truth." The project focuses on a professional, trustworthy design that simplifies the complex process of finding doctors and understanding medical conditions. It addresses the needs of three distinct groups: patients seeking care, doctors managing practice, and researchers seeking data.

3 Problem Definition

- **Misinformation:** Difficulty in finding credible, structured information on diseases.
- **Accessibility:** Rural or busy users struggle to find and book specialized doctors quickly.
- **Data Fragmentation:** Lack of a unified space where patients can store medical history and researchers can access datasets.
- **Navigation Fatigue:** Users often feel overwhelmed by complex medical portals.

4 Objectives

- **Intuitive UX:** A clean, professional interface focusing on simplicity and trust to reduce user anxiety.
- **Credible Encyclopedia:** An articles section containing symptoms, causes, and treatments sourced from global health organizations like WHO/CDC.
- **AI-Powered Patient Assistance:** A specialized **Chatbot for Patients** to act as a 24/7 concierge, helping them find specific disease articles, navigate the platform, or understand how to book an appointment.
- **Efficient Search:** Advanced filtering by category, keyword, or disease name for quick information retrieval.
- **Secure Access:** Role-based authentication (RBAC) via Firebase for Patients, Doctors, and Admins.
- **Appointment Lifecycle:** End-to-end management from initial scheduling to payment and medical history tracking.
- **Research Support:** A dedicated module for uploading and managing hospital datasets to support the medical research community.

5 System Analysis

Functional Requirements

- **Admin Panel:** Full CRUD (Create, Read, Update, Delete) control over users, articles, and appointment logs.
- **Doctor Dashboard:** Real-time visibility of schedules and patient history access.
- **Patient Dashboard:** A private vault to upload and manage personal medical records.
- **AI Navigation:** A Chatbot powered by Gemini API to guide users to relevant articles.

Non-Functional Requirements

- **Security:** Data encryption via secure REST API endpoints.
- **Reliability:** High accuracy of article content.

6 System Design:

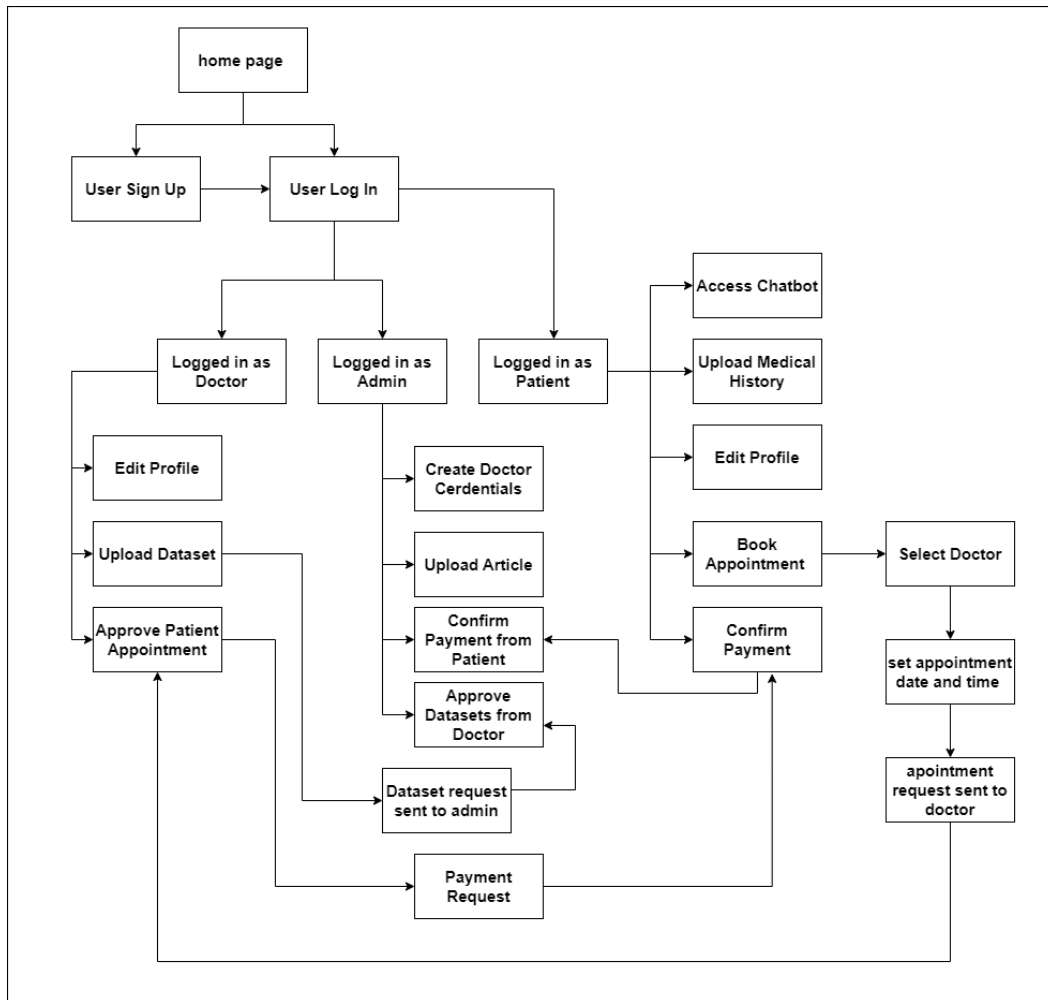


Figure 1 : Flowchart of Overall Project

7 Technology Stack

- **Frontend:** ReactJS (Single Page Application for speed and fluid navigation).
- **Backend:** Node.js & ExpressJS (Handling business logic and API routing).
- **Database:** MongoDB (for flexible article content and chat history) or SQL (for structured appointment/user data).
- **Authentication:** JWT
- **Intelligence:** Gemini API (Natural Language Processing for the Chatbot).

8 Database Design

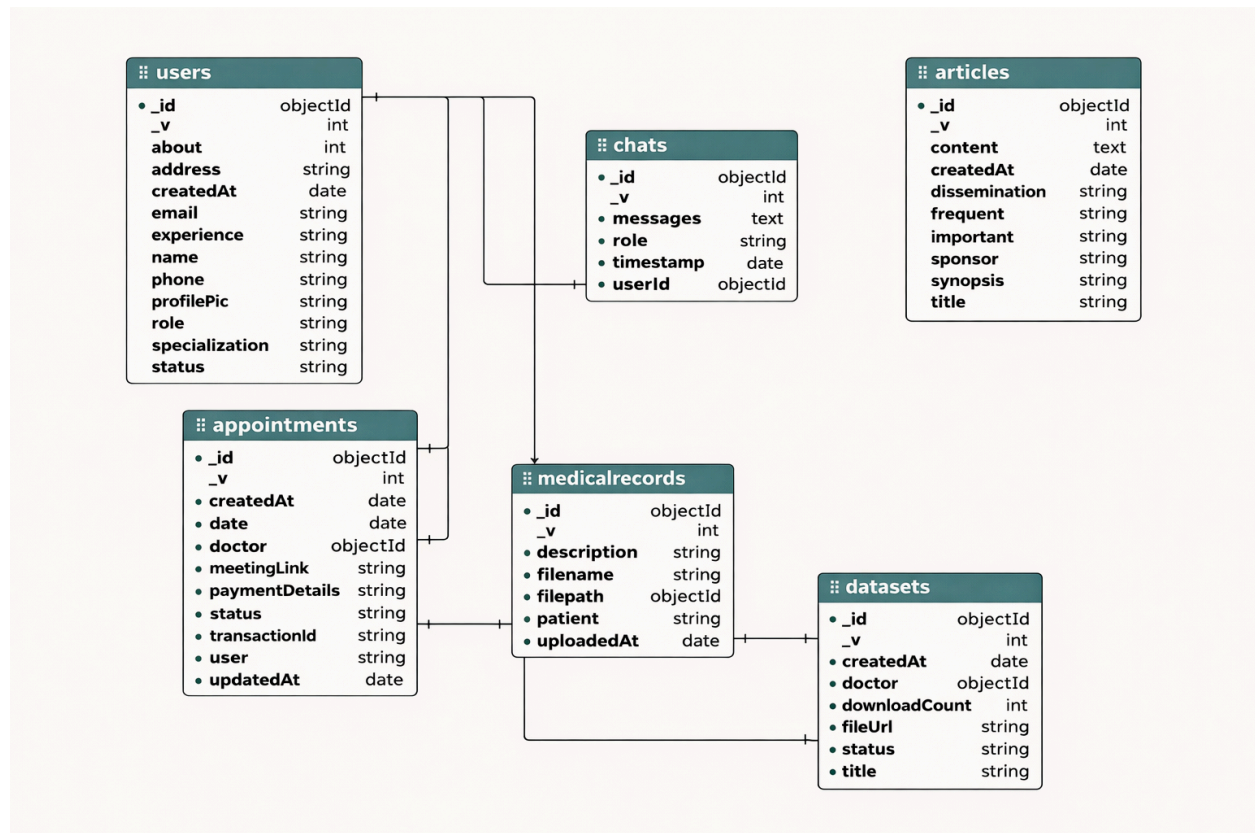


Figure 2 : Database Design generated by mongodb data model

9 Screenshots of Project

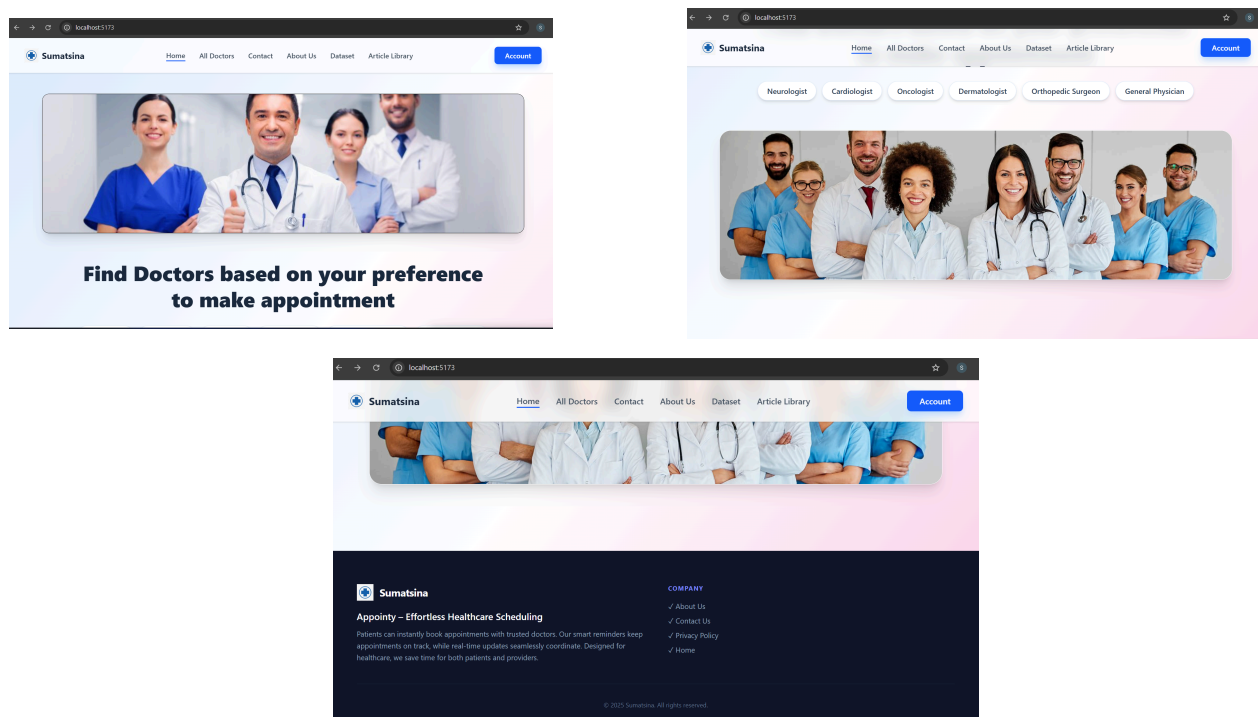


Figure 3: Home page with Header and footer

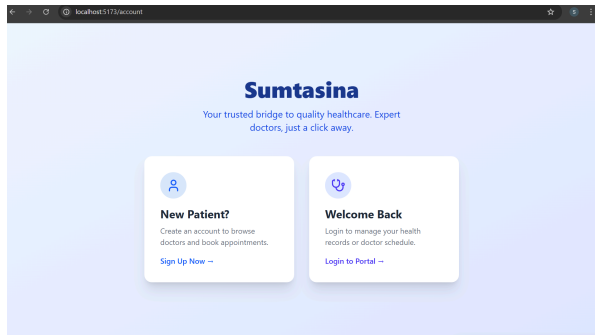


Figure 4: Account page

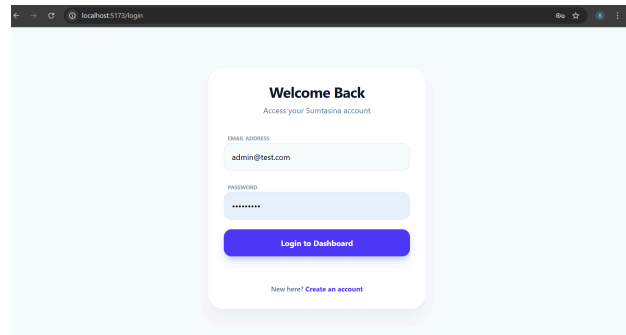


Figure 5: Login page

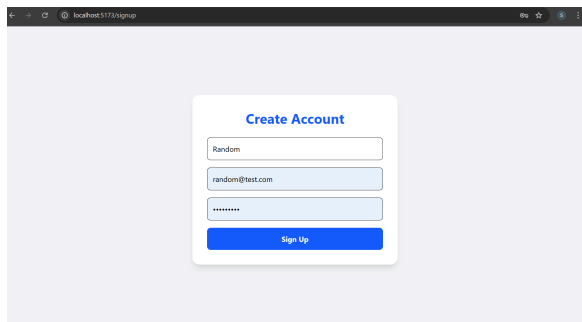


Figure 6: Create Account Page

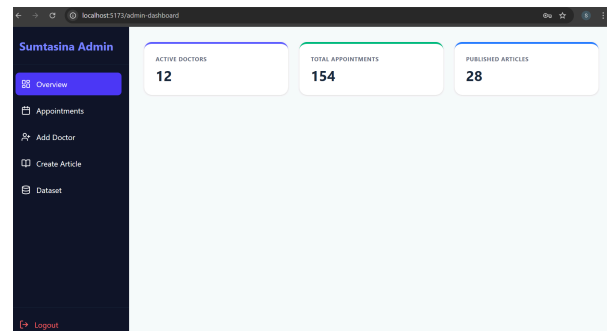


Figure 7: Admin Dashboard

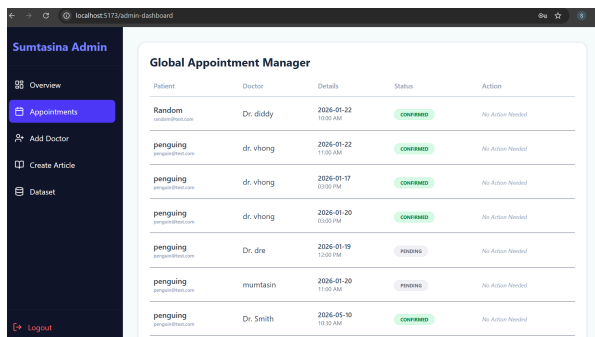


Figure 8: Admin Appointment manager

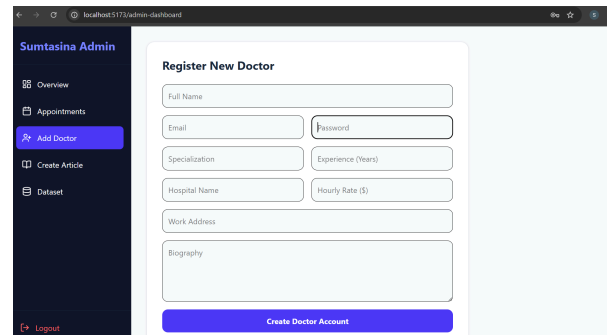


Figure 9: Admin registering new doctor

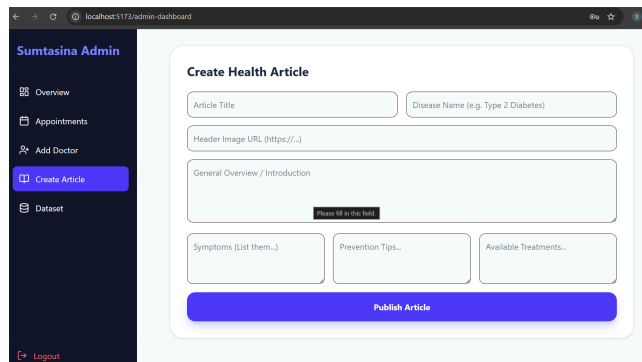


Figure 10: Create Health Article

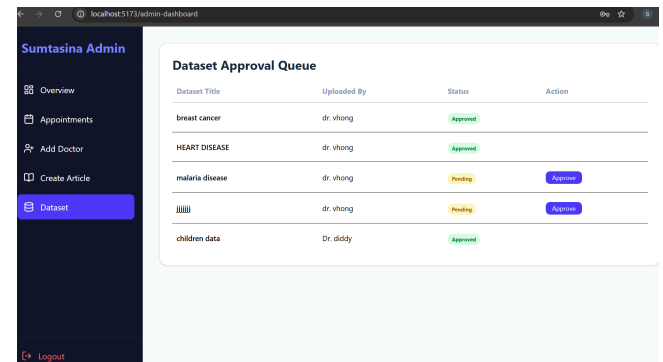


Figure 11: Admin Dataset Manager

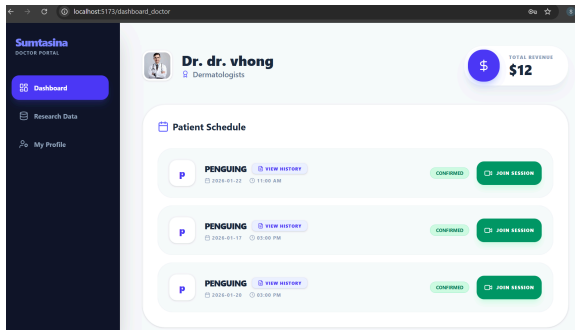


Figure 12 : Doctor Dashboard

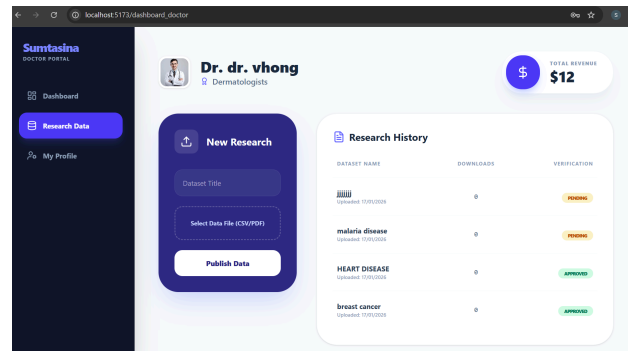


Figure 13 : Dataset upload page

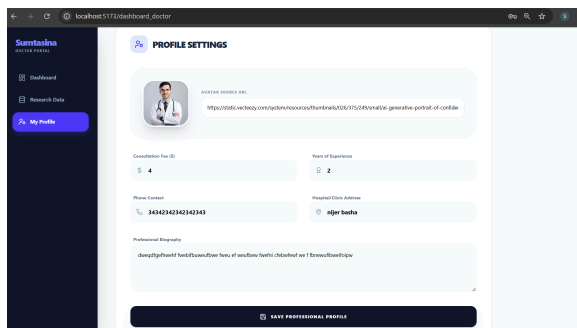


Figure 14 : Doctor edit profile page

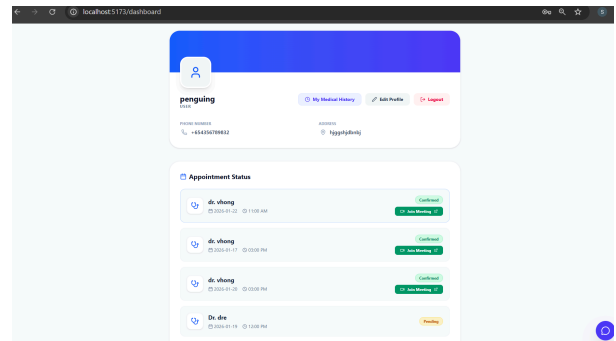


Figure 15 : Patient Dashboard

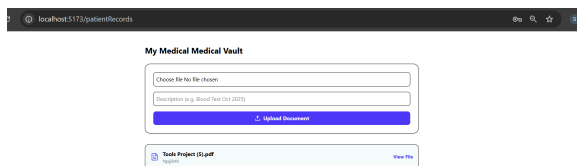


Figure 16 : Medical history upload & history

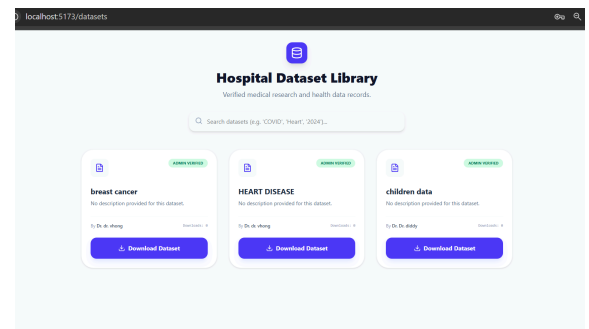


Figure 17 : Hospital dataset Library

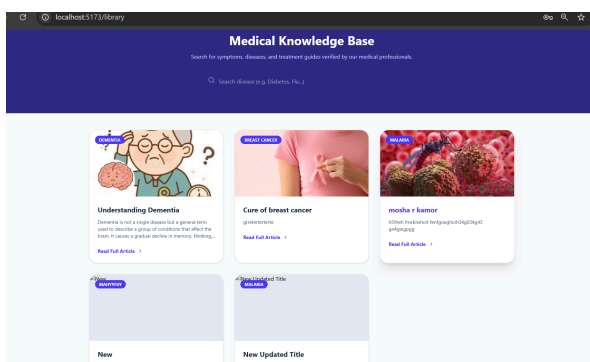


Figure 18 : Article Library

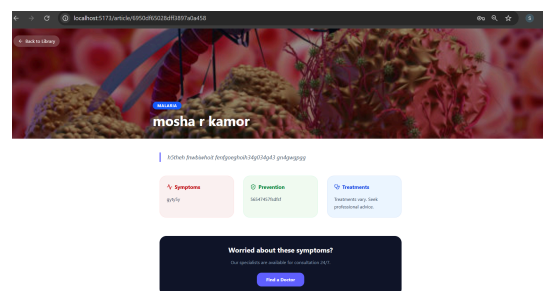


Figure 17 : Article Details

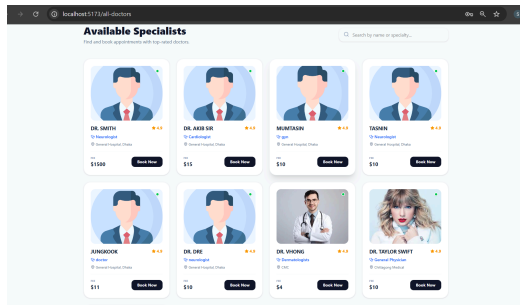


Figure 18: All doctor Page

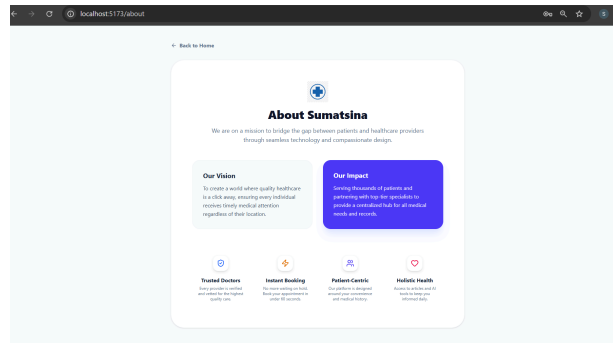


Figure 19 : AboutUs

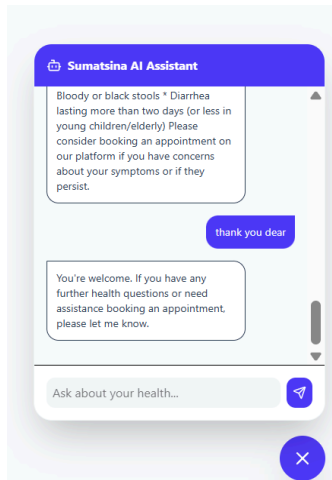


Figure 20 : Chatbot

9 Key Features (Summary)

1. **Dual Dashboards:** Custom views for healthcare providers and recipients.
2. **Medical Vault:** Patient-controlled medical history management.
3. **Encyclopedia Mode:** Search/Filter functionality for medical literacy.
4. **AI Assistant:** Intelligent site navigation and article discovery.
5. **Research Hub:** Hospital dataset portal for academic and clinical research.

10 Limitations & Future Scope

- **Limitations:** Initial dependence on manual dataset uploads; AI chatbot requires continuous fine-tuning for medical terminology.
- **Future Scope:** Integration of Telemedicine (Video calls), automated AI diagnosis suggestions, and IoT integration for real-time patient vitals tracking.

11 Conclusion

Sumatsina successfully creates a reliable, digital healthcare ecosystem. It is more than just a booking tool; it is an educational resource and a research facilitator. By focusing on user experience and data credibility, the project ensures that digital healthcare is accessible, secure, and deeply informative for all stakeholders.

